Documentation Serveur BDD

1. Introduction

Ce document decrit la structure de la base de donnees utilisee par l'application web. Il a ete genere a partir d'un dump SQLite (Adminer 4.8.4) et recense l'ensemble des tables, leurs colonnes, leurs types, leurs contraintes ainsi que le code SQL de creation.

La base de donnees est composee de deux categories de tables :

- Tables metier : users, appointments, clients, products, purchases, audit_logs, personal_access_tokens.
- Tables systeme Laravel : cache, cache_locks, jobs, job_batches, failed_jobs, sessions, password_reset_tokens, migrations.

2. Informations generales

SGBD	SQLite 3.49.2
Outil de dump	Adminer 4.8.4
Framework	Laravel (PHP)
Encodage	UTF-8
Nombre de tables metier	6
Nombre de tables systeme	8
Cle primaire par defaut	INTEGER PRIMARY KEY AUTOINCREMENT

3. Schema des tables metier

Table : users

Colonne	Type	Contrainte	Description
`id`	INTEGER	**PK / AUTO**	Identifiant unique de l'utilisateur
`name`	VARCHAR	NOT NULL	Nom complet de l'utilisateur
`email`	VARCHAR	UNIQUE	Adresse email (index unique)
`email_verified_at`	DATETIME	NULL	Date de verification de l'email
`password`	VARCHAR	NOT NULL	Mot de passe hache (bcrypt)
`remember_token`	VARCHAR	NULL	Token de connexion persistante
`created_at`	DATETIME	NULL	Date de creation de l'enregistrement
`updated_at`	DATETIME	NULL	Date de derniere modification

Table : appointments

Colonne	Type	Contrainte	Description
`id`	INTEGER	**PK / AUTO**	Identifiant unique du rendez-vous
`user_id`	INTEGER	NOT NULL	Utilisateur concerne (FK -> users)
`client_name`	VARCHAR	NOT NULL	Nom du client pour ce rendez-vous
`starts_at`	DATETIME	NOT NULL	Date et heure de debut du rendez-vous
`notes`	TEXT	NULL	Notes ou commentaires libres
`created_at`	DATETIME	NULL	Date de creation
`updated_at`	DATETIME	NULL	Date de modification

Cles etrangeres : `user_id -> users(id) ON DELETE CASCADE`

Table : clients

Colonne	Type	Contrainte	Description
`id`	INTEGER	**PK / AUTO**	Identifiant unique du client
`name`	VARCHAR	NOT NULL	Raison sociale ou nom du client
`email`	VARCHAR	NULL	Adresse email du client
`phone`	VARCHAR	NULL	Numero de telephone
`created_at`	DATETIME	NULL	Date de creation
`updated_at`	DATETIME	NULL	Date de modification

Table : products

Colonne	Type	Contrainte	Description
`id`	INTEGER	**PK / AUTO**	Identifiant unique du produit
`name`	VARCHAR	NOT NULL	Nom commercial du produit
`price`	NUMERIC	DEFAULT 0	Prix unitaire HT
`description`	TEXT	NULL	Description detaillee du produit
`created_at`	DATETIME	NULL	Date de creation
`updated_at`	DATETIME	NULL	Date de modification

Table : purchases

Colonne	Type	Contrainte	Description
`id`	INTEGER	**PK / AUTO**	Identifiant unique de l'achat
`user_id`	INTEGER	NOT NULL	Commercial responsable (FK -> users)
`client_id`	INTEGER	NOT NULL	Client acheteur (FK -> clients)
`product_id`	INTEGER	NOT NULL	Produit achete (FK -> products)
`quantity`	INTEGER	DEFAULT 1	Quantite achetee
`purchased_at`	DATETIME	NOT NULL	Date effective de l'achat
`notes`	VARCHAR	NULL	Observations eventuelles
`created_at`	DATETIME	NULL	Date de creation
`updated_at`	DATETIME	NULL	Date de modification

Cles etrangeres : `user_id -> users(id) CASCADE | client_id -> clients(id) CASCADE | product_id -> products(id) CASCADE`

Table : audit_logs

Colonne	Type	Contrainte	Description
id	INTEGER	**PK / AUTO**	Identifiant unique du log
user_id	INTEGER	NULL	Utilisateur ayant declenche l'action (FK -> users)
action	VARCHAR	NOT NULL	Libelle de l'action effectuee
entity_type	VARCHAR	NOT NULL	Type d'entite concernee (ex: App\Models\User)
entity_id	INTEGER	NULL	Identifiant de l'entite concernee
meta	TEXT	NULL	Donnees supplementaires (JSON)
created_at	DATETIME	NULL	Date de creation
updated_at	DATETIME	NULL	Date de modification

Cles etrangeres : `user_id -> users(id) ON DELETE SET NULL`

Table : personal_access_tokens

Colonne	Type	Contrainte	Description
id	INTEGER	**PK / AUTO**	Identifiant unique du token
tokenable_type	VARCHAR	NOT NULL	Type du modele proprietaire (polymorphe)
tokenable_id	INTEGER	NOT NULL	ID du proprietaire du token
name	TEXT	NOT NULL	Libelle du token (ex: postman)
token	VARCHAR	UNIQUE	Valeur du token hachee (SHA-256)
abilities	TEXT	NULL	Permissions accordees (JSON)
last_used_at	DATETIME	NULL	Derniere utilisation du token
expires_at	DATETIME	NULL	Date d'expiration (NULL = indefini)
created_at	DATETIME	NULL	Date de creation
updated_at	DATETIME	NULL	Date de modification

4. Tables systeme Laravel

Les tables suivantes sont generees et gerees automatiquement par le framework Laravel. Elles ne doivent pas etre modifiees manuellement.

cache	Stockage du cache applicatif cle/valeur avec TTL
cache_locks	Verrous distribues pour les operations atomiques sur le cache
jobs	File d'attente des taches asynchrones (queues)
job_batches	Suivi des lots de taches (batch jobs)
failed_jobs	Journalisation des taches ayant echoue
sessions	Sessions utilisateurs (id, IP, user_agent, payload)

password_reset_tokens	Tokens temporaires pour la reinitialisation de mot de passe
migrations	Historique des migrations de schema executees

5. Relations et index

Le tableau ci-dessous recapitule les cles etrangeres et les index presents dans le schema.

Table	Index / FK	Colonnes	Comportement
users	UNIQUE INDEX	email	Unicite de l'email
appointments	FK	user_id -> users	ON DELETE CASCADE
purchases	FK	user_id -> users	ON DELETE CASCADE
purchases	FK	client_id -> clients	ON DELETE CASCADE
purchases	FK	product_id -> products	ON DELETE CASCADE
purchases	INDEX	(user_id, client_id)	Optimisation des requetes
audit_logs	FK	user_id -> users	ON DELETE SET NULL
audit_logs	INDEX	(user_id, action)	Recherche par action
audit_logs	INDEX	(entity_type, entity_id)	Recherche par entite
personal_access_tokens	UNIQUE INDEX	token	Unicite du token
personal_access_tokens	INDEX	expires_at	Nettoyage des tokens expir.
personal_access_tokens	INDEX	(tokenable_type, id)	Acces polymorphe
sessions	INDEX	last_activity	Expiration des sessions
sessions	INDEX	user_id	Recherche par utilisateur
failed_jobs	UNIQUE INDEX	uuid	Deduplication

6. Code SQL de creation

Le code SQL suivant permet de recreer integralement le schema de la base de donnees (sans les donnees). Les tables sont ordonnees pour respecter les dependances de cles etrangeres.

6.1 Table users

```
CREATE TABLE "users" (
    "id"                INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "name"              VARCHAR NOT NULL,
    "email"             VARCHAR NOT NULL,
    "email_verified_at" DATETIME,
    "password"          VARCHAR NOT NULL,
```

```
    "remember_token"    VARCHAR,
    "created_at"        DATETIME,
    "updated_at"        DATETIME
);

CREATE UNIQUE INDEX "users_email_unique" ON "users" ("email");
```

6.2 Table appointments

```
CREATE TABLE "appointments" (
    "id"           INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "user_id"      INTEGER NOT NULL,
    "client_name" VARCHAR NOT NULL,
    "starts_at"    DATETIME NOT NULL,
    "notes"        TEXT,
    "created_at"   DATETIME,
    "updated_at"   DATETIME,
    FOREIGN KEY ("user_id") REFERENCES "users"("id") ON DELETE CASCADE
);
```

6.3 Table clients

```
CREATE TABLE "clients" (
    "id"         INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "name"       VARCHAR NOT NULL,
    "email"      VARCHAR,
    "phone"      VARCHAR,
    "created_at" DATETIME,
    "updated_at" DATETIME
);
```

6.4 Table products

```
CREATE TABLE "products" (
    "id"           INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "name"         VARCHAR NOT NULL,
    "price"        NUMERIC NOT NULL DEFAULT '0',
    "description" TEXT,
    "created_at"   DATETIME,
    "updated_at"   DATETIME
);
```

6.5 Table purchases

```
CREATE TABLE "purchases" (
    "id"            INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "user_id"       INTEGER NOT NULL,
    "client_id"     INTEGER NOT NULL,
    "product_id"    INTEGER NOT NULL,
```

```
    "quantity"     INTEGER NOT NULL DEFAULT '1',
    "purchased_at" DATETIME NOT NULL,
    "notes"        VARCHAR,
    "created_at"   DATETIME,
    "updated_at"   DATETIME,
    FOREIGN KEY ("user_id")    REFERENCES "users"("id")    ON DELETE CASCADE,
    FOREIGN KEY ("client_id")  REFERENCES "clients"("id")  ON DELETE CASCADE,
    FOREIGN KEY ("product_id") REFERENCES "products"("id") ON DELETE CASCADE
);

CREATE INDEX "purchases_user_id_client_id_index"
    ON "purchases" ("user_id", "client_id");
```

6.6 Table audit_logs

```
CREATE TABLE "audit_logs" (
    "id"          INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "user_id"     INTEGER,
    "action"      VARCHAR NOT NULL,
    "entity_type" VARCHAR NOT NULL,
    "entity_id"   INTEGER,
    "meta"        TEXT,
    "created_at"  DATETIME,
    "updated_at"  DATETIME,
    FOREIGN KEY ("user_id") REFERENCES "users"("id") ON DELETE SET NULL
);

CREATE INDEX "audit_logs_user_id_action_index"
    ON "audit_logs" ("user_id", "action");
CREATE INDEX "audit_logs_entity_type_entity_id_index"
    ON "audit_logs" ("entity_type", "entity_id");
```

6.7 Table personal_access_tokens

```
CREATE TABLE "personal_access_tokens" (
    "id"             INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "tokenable_type" VARCHAR NOT NULL,
    "tokenable_id"   INTEGER NOT NULL,
    "name"           TEXT NOT NULL,
    "token"          VARCHAR NOT NULL,
    "abilities"      TEXT,
    "last_used_at"   DATETIME,
    "expires_at"     DATETIME,
    "created_at"     DATETIME,
    "updated_at"     DATETIME
);

CREATE UNIQUE INDEX "personal_access_tokens_token_unique"
    ON "personal_access_tokens" ("token");
CREATE INDEX "personal_access_tokens_expires_at_index"
```

```
    ON "personal_access_tokens" ("expires_at");
CREATE INDEX "personal_access_tokens_tokenable_type_tokenable_id_index"
    ON "personal_access_tokens" ("tokenable_type", "tokenable_id");
```

6.8 Tables systeme Laravel

```
CREATE TABLE "cache" (
    "key"        VARCHAR NOT NULL,
    "value"      TEXT NOT NULL,
    "expiration" INTEGER NOT NULL,
    PRIMARY KEY ("key")
);

CREATE TABLE "cache_locks" (
    "key"        VARCHAR NOT NULL,
    "owner"      VARCHAR NOT NULL,
    "expiration" INTEGER NOT NULL,
    PRIMARY KEY ("key")
);

CREATE TABLE "jobs" (
    "id"           INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "queue"        VARCHAR NOT NULL,
    "payload"      TEXT NOT NULL,
    "attempts"     INTEGER NOT NULL,
    "reserved_at"  INTEGER,
    "available_at" INTEGER NOT NULL,
    "created_at"   INTEGER NOT NULL
);
CREATE INDEX "jobs_queue_index" ON "jobs" ("queue");

CREATE TABLE "failed_jobs" (
    "id"         INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,
    "uuid"       VARCHAR NOT NULL,
    "connection" TEXT NOT NULL,
    "queue"      TEXT NOT NULL,
    "payload"    TEXT NOT NULL,
    "exception"  TEXT NOT NULL,
    "failed_at"  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE UNIQUE INDEX "failed_jobs_uuid_unique" ON "failed_jobs" ("uuid");

CREATE TABLE "sessions" (
    "id"            VARCHAR NOT NULL,
    "user_id"       INTEGER,
    "ip_address"    VARCHAR,
    "user_agent"    TEXT,
    "payload"       TEXT NOT NULL,
    "last_activity" INTEGER NOT NULL,
    PRIMARY KEY ("id")
```

```
);
CREATE INDEX "sessions_last_activity_index" ON "sessions" ("last_activity");
CREATE INDEX "sessions_user_id_index"        ON "sessions" ("user_id");

CREATE TABLE "password_reset_tokens" (
    "email"      VARCHAR NOT NULL,
    "token"      VARCHAR NOT NULL,
    "created_at" DATETIME,
    PRIMARY KEY ("email")
);
```